

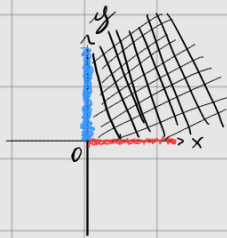
Função com duas variáveis

$$f(x,y) = \sqrt{x} + \sqrt{y}$$

Restrições: $x \geq 0$ e $y \geq 0$

$$D = \{(x,y) \in \mathbb{R}^2 \mid x \geq 0 \text{ e } y \geq 0\}$$

$$\text{Im}(f) = \mathbb{R}_+$$



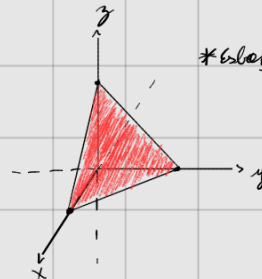
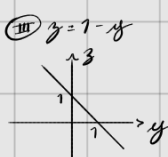
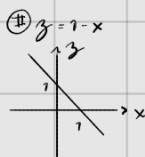
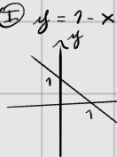
* Deve-se sempre representar o domínio das funções antes de tentar esboçar gráficos.

$$f(x,y) = 1 - x - y$$

* Planos: $\text{I} \ x,y$ ou $\text{II} \ x,z$ ou $\text{III} \ y,z$

$$D(f) = \mathbb{R} \quad \text{Im}(f) = \mathbb{R}$$

* Não há restrições



* Esboço do gráfico da função

$$f(x,y) = x^2 + y^2$$

$$D(f) = \mathbb{R}^2$$

$$\text{Im} = [0, +\infty)$$

$$z = x^2 + y^2$$

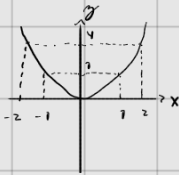
Plano $x,y \mid z=0$

$$x^2 + y^2 = 0 \Rightarrow |x,y| = (0,0)$$



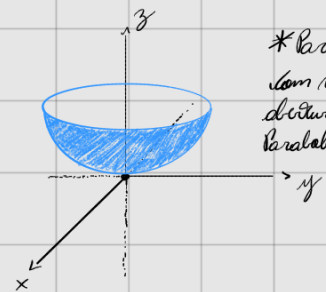
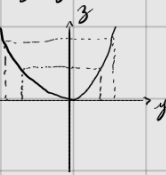
Plano $x,z \mid y=0$

$$z = x^2 + 0$$



Plano $y,z \mid x=0$

$$z = y^2$$



* Parabolóide com vértice $(0,0,0)$ abertura p/ cima Parabolóide Circular

("Para todo y", mesmo que não delimitado)

$$f(x,y) = 1 - x^2$$

$$D(f) = \mathbb{R}^2$$

$$\text{Im}(f) = [-\infty, 1]$$

$$z = 1 - x^2$$

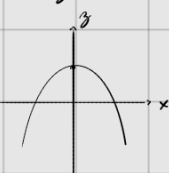
Plano $x,y \mid z=0$

$$1 - x^2 = 0 \Rightarrow x = 1 \text{ e } x = -1$$



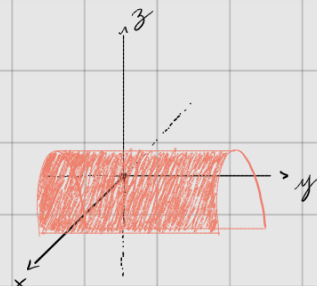
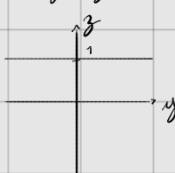
Plano $x,z \mid y=0$

$$1 - x^2 = z$$



Plano $y,z \mid x=0$

$$1 - x^2 = z \Rightarrow z = 1$$



$$f(x,y) = \sqrt{x^2 + y^2}$$

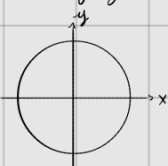
$$D(f) = \mathbb{R}^2$$

$$\text{Im}(f) = \mathbb{R}_+$$

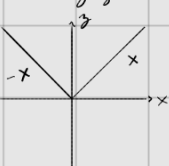
$$x^2 + y^2 \geq 0$$

* Sempre satisfato

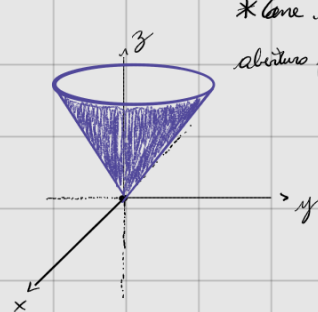
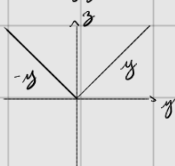
Plano $x,y \mid z=0$



Plano $x,z \mid y=0$



Plano $y,z \mid x=0$



* Cone Circular com V(0,0,0)

abertura p/ cima com projeção em $\mathbb{R}_+$

Curvas de nível:

$$z = 1 \Rightarrow x^2 + y^2 = 1$$

$$z = 2 \Rightarrow x^2 + y^2 = 4$$

$$z = 3 \Rightarrow x^2 + y^2 = 9$$

$$0 = \sqrt{x^2 + y^2}$$

$$x^2 + y^2 = 0$$

$$z = \sqrt{x^2} = |x|$$

$$|x| \begin{cases} x, & \text{se } x \geq 0 \\ -x, & \text{se } x < 0 \end{cases}$$

$$z = \sqrt{y^2} = |y|$$

$$|y| \begin{cases} y, & \text{se } y \geq 0 \\ -y, & \text{se } y < 0 \end{cases}$$

* As curvas de nível demonstram o comportamento de abertura do cone, pois abrem-se que no plano x,y é circunferência aumento de raio